

Pricing Sovereign Credit Risk in Oil-Exporting Economies

MSc thesis exposé

Motivation & Literature Gap

Commodity prices are a recognized driver of credit risk in emerging market economies. In the case of oil price risk, there is literature documenting its statistical relationship with sovereign CDS spreads, and separate work showing how oil revenues affect national balance sheets and fiscal capacity. However, this knowledge has not been brought into structural default models, which provide a stronger economic intuition as to why defaults happen.

Literature Stream	Key Papers	What It Establishes
Structural models for sovereigns	Gray, Merton & Bodie (2007) Gapen et al. (2005, 2008) Duyvesteyn & Martens (2015)	Structural framework treating sovereign balance sheet as options. Equity (local currency claims) as junior claim, USD denominated debt as the senior claim. Default is endogenous, and the observed market CDS should correlate with the model Distance-to-distress (the number of standard deviations the sovereign asset value is away from the debt barrier)
Oil–Sovereign Transmission	Bouri et al. (2018, 2019) Wegener et al. (2025) Naifar et al. (2020) Sabkha et al. (2019)	Oil prices affect sovereign CDS spreads. The relationship is nonlinear, asymmetric, and regime-dependent (stronger during stress), different effect on exporters. However the models used here are reduced-form, which lack the economic intuition of structural models.
Jump-Diffusion Credit Models	Zhou (1997, 2001) Wong & Li (2006) Da Fonseca et al. (2016)	Jumps capture sudden movements in asset value that smooth diffusion misses, and help better explain spreads, but have been applied to corporate debt only.

Research gap

There is no research as to how the oil price dynamics could extend structural models. This gap matters because it leaves us without an economic mechanism connecting commodity prices to sovereign credit risk. This thesis would provide a structural specification for the transmission channels of oil price risk into sovereign debt.

Empirical Evidence Framework

Empirical evidence that supports the choice of model and the proposed extension

I. Oil price risk transmits differently to exporters than importers

Main motivation

- a Correlation between oil returns and bond/equity returns in each country
- b Oil doesn't affect exporter CDS spreads all the time, only when there is high implied volatility in oil. We run a panel regression of CDS changes on oil returns $\times$ exporter dummy. The interaction coefficient goes from insignificant to highly significant as we move up in volatility regime.
- c When oil is relatively expensive, exporters show lower relative CDS spreads. This doesn't show for non-exporters

II. Exchange rates of exporters are suppressed

Motivates model extension 1 (FX adjustment)

- d Equity/Bond markets co-move with oil returns in exporters, but FX rates doesn't.
- e The residual from regressing FX returns on Oil returns explains future FX returns

III. Oil and CDS series exhibit jump behavior

Motivates model extension 2 (jump-diffusion)

- f Oil and CDS contain sudden large moves, that aren't explained by volatility clustering alone. A GARCH-Jump model fits significantly better by BIC and log-likelihood than a GARCH only model
- g Jumps in Oil and CDS are skewed (negative for oil, positive for CDS) and tend to happen at the same time

Baseline model: Contingent Claims Analysis

Gray, Merton & Bodie (2007); Gapen et al. (2005, 2008)

Sovereign Balance Sheet

ASSETS	LIABILITIES
Foreign Reserves	Foreign Currency Debt <i>= Distress Barrier B</i>
Net Fiscal Assets	
Other Public Assets	Local Currency Liabilities
	Monetary Base
	Domestic Debt
 <i>Implied sovereign asset value A</i> <i>(unobservable)</i>	 <i>Equity claim</i> <i>(observable)</i>

Contingent Claims Interpretation

Local Currency Liabilities = **Junior Claim** = **Call Option on A**

$$LCL = \max(A - B, 0)$$

Foreign Currency Debt = **Risk-Free Debt** – **Put Option**

$$FC\ Debt = B \cdot e^{-rT} - P(A, B, \sigma_A, r, T)$$

Sovereign Credit Risk = Value of the Implicit Put Option

(1) Value equation:

$$LCL\ (USD) = A \cdot N(d_1) - B \cdot e^{-rT} \cdot N(d_2)$$

$$d_1 = [\ln(A/B) + (r + \frac{1}{2}\sigma_A^2)T] / (\sigma_A \sqrt{T})$$

$$d_2 = d_1 - \sigma_A \sqrt{T}$$

(2) Volatility transfer:

$$\sigma_{LCL} \cdot LCL = N(d_1) \cdot \sigma_A \cdot A$$

Proposed Model Extensions & Testing

Extension 1: Oil-Adjusted Exchange Rate

Motivated by Category II (evidence d, e)

What it captures:

The baseline CCA converts local currency liabilities into USD at the spot rate. We let the exporter currencies become more or less valuable with the price of oil, which makes the equity claim reflect oil price dynamics

Mechanism:

Oil high → adjusted FX stronger → LCL(USD) larger → equity more valuable → lower risk

Oil low → adjusted FX weaker → LCL(USD) smaller → equity less valuable → higher risk

Extension 2: Jump-Diffusion Asset Dynamics

Motivated by Category III (evidence f, g)

What it captures:

The baseline CCA uses geometric Brownian motion (continuous). This extension adds sudden negative shocks. The sovereign asset value can drop suddenly.

Calibration:

Jump intensity (λ) linked to OVX — high oil volatility → more frequent jumps. → higher risk during high stress regimes

How we test: three-model comparison

Model A: Baseline CCA



Model B: CCA + FX adjustment



Model C: CCA + jump-adjustment

Each model produces a distance to distress metric (measure of how many stdev is the sovereign asset value away from the distress barrier). We compare them against market CDS spreads using correlation and R^2 . The model extensions should produce higher R^2 in exporters but produce no changes in importers.